ELECTRONIC SPEED CONTROLLER

ARK 4IN1 ESC

NDAA-compliant 4-in-1 ESC running ARK32 firmware

The USA-built ARK 4IN1 ESC is a high-performance 4-channel electronic speed controller for brushless motors. Four independent STM32F051 MCUs with TI DRV8328 gate drivers drive four 3-phase power stages, delivering 50 A continuous / 75 A burst per motor under ARK32 firmware (supports AM32).

A 0.2 mΩ shunt shared by all four channels senses total battery current at 100 A/V and limits the board to about 170 A continuous. Configure KV and pole count for each motor before first use.

KEY FEATURES

- 50 A continuous / 75 A burst per motor
- 3–8S LiPo (6 V min, 65 V abs max)
- 4× independent STM32F051 channels
- Onboard current sensor (100 A/V)
- ARK32 firmware (supports AM32)
- DShot 300/600, bidirectional DShot, KISS telemetry, PWM
- 30.5 mm mounting pattern

SPECIFICATIONS

ELECTRICAL	
Input voltage	3–8S LiPo, 6 V minimum
Absolute maximum	65 V
Continuous current	50 A per motor, with airflow
Burst current	75 A per motor
Little airflow (hover)	25–30 A per motor
Board total	About 170 A continuous
Bulk capacitance	450 µF
Current sense	100 A/V
Firmware	ARK32 (supports AM32)
Operating temperature	−25 °C to +85 °C
Storage temperature	−25 °C to +85 °C

MECHANICAL	
Dimensions	43.0 × 40.5 × 7.6 mm
Weight	14.5 g
Mounting	30.5 mm pattern



AT A GLANCE	
Voltage	3–8S LiPo
Current / motor	50 A cont / 75 A burst
Firmware	ARK32 (AM32)
MCU	4× STM32F051
Sense	100 A/V
Dimensions	43 × 40.5 × 7.6 mm
Weight	14.5 g
Hardware rev	Rev 4.2

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework listed

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-4in1-esc/

DOCS docs.arkelectron.com/products/electronic-speed-controller/ark-4in1-esc

CAD github.com/ARK-Electronics/ark-hardware

FLIGHT CONTROLLER CONNECTOR

PIN	SIGNAL	NOTES
1	VBAT	Battery voltage
2	CURR	Current sense output
3	TELEM	Serial telemetry
4	MOTOR 1	DShot / PWM
5	MOTOR 2	DShot / PWM
6	MOTOR 3	DShot / PWM
7	MOTOR 4	DShot / PWM
8	GND	

Connector: 8-pin JST-SH. Matches the ARK FPV ESC connector through the included cable.

DEBUG CONNECTOR

PIN	SIGNAL	NOTES
1	3.3 V	
2–9	SWDIO / SWDCLK 1–4	Per-motor SWD
10	GND	

Connector: 10-pin JST-SH.

NOTES

- Before first use, configure KV and pole count for each motor.
- DShot is the recommended input protocol. PWM requires calibration.
- Motor and battery connections are solder pads.
- Current ratings are board thermal limits with airflow and have no fixed duration. All four channels share one current-sense shunt, which limits the board to about 170 A, or about 43 A per motor with all four loaded.
- The 450 µF of onboard bulk capacitance covers 3S–6S with short battery leads. At 8S, add an external bulk capacitor across the battery input to keep transients below the 65 V absolute maximum.
- Includes 4 vibration dampers and the ARK FPV cable.

VARIANT

PRODUCT	KEY DIFFERENCE
ARK 4IN1 ESC CONS	Connectorized (MR30 motors, battery cables); 30 A cont / 40 A burst limited by connectors; larger size / weight

ARK-DS-ARK-4IN1-ESC | Version 1.0, 2026-10-29 | Compiled from the ARK 4IN1 ESC Production schematic Rev 4.2 (28 May 2026), the Production bill of materials, manufacturing documentation, and ARK Electronics published documentation. Current ratings are board thermal limits. Temperature ratings per component data sheets; the JST-SH connectors set −25 °C. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.